Module 6 Project – Signature Assessment

ALY6130: Risk Management Analytics

Academic Term: Winter Term 2026

Module: Module 6

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Final Project - Signature Assessment

1. Background

Fraser Health Authority (FHA) is one of British Columbia's largest publicly funded regional health authorities, serving a population of over 1.9 million residents across the Lower Mainland and Fraser Valley. FHA operates a comprehensive network of hospitals, community health centres, and specialized services, delivering care across a wide geographic and demographic spectrum. In recognition of evolving healthcare delivery models and growing demand for home-based and virtual care, Fraser Health identified a strategic opportunity in the acquisition and deployment of a Remote Patient Monitoring (RPM) service platform.

RPM technology allows healthcare providers to remotely monitor patients using connected medical devices and digital platforms, facilitating real-time data collection on vital signs, chronic conditions, and patient-reported outcomes. The evidence base for RPM is compelling: research demonstrates that such platforms reduce hospital readmission rates, improve chronic disease management outcomes, and enhance patient engagement in their own care (Keesara et al., 2020; Shah et al., 2018). However, the expansion of digital health infrastructure does not occur in a risk-free environment. Healthcare technology adoption introduces cybersecurity vulnerabilities, regulatory compliance obligations, operational integration challenges, and competitive dynamics that organizations must proactively evaluate and manage.

This Signature Assessment aggregates and synthesizes the full body of enterprise risk management (ERM) analysis conducted throughout the ALY6130 course term. The work spans five sequential modules, each building upon the last: Module 1 established a SWOT analysis identifying Fraser Health's internal strengths and weaknesses, and external opportunities and threats associated with RPM acquisition. Module 2 identified three focal risks, defined them using structured risk statements, and established early Key Risk Indicators (KRIs). Module 3 conducted quantitative risk scoring using a risk calculation sheet, risk register, and probability-impact heat map, supplemented by machine learning classification. Module 4 applied the Indicators and Warning (I&W) analytical technique for quantitative risk assessment, validated by Monte Carlo simulation and Random Forest classification. The final deliverable, this report synthesizes all work into a comprehensive ERM report suitable for senior decision-makers.

The analytical framework applied throughout this assessment is grounded in the Committee of Sponsoring Organizations of the Treadway Commission (COSO, 2017) Enterprise Risk Management framework, which integrates risk management with organizational strategy and performance. This approach ensures that risks are evaluated not in isolation, but in the context of Fraser Health's mission, strategic objectives, and operational environment.

2. Identification of Risks

Risk identification for Fraser Health's RPM platform acquisition was conducted through a structured, multi-method approach. Beginning with the SWOT analysis in Module 1, the team evaluated internal capabilities and external environmental factors to surface potential risk domains. From this foundation, 3 focal risks were identified.

Each risk was articulated using the structured format: 'The risk of… caused by… resulting in…' to ensure clarity, specificity and actionability in the risk statements. The Risk Register captures each risk's description, likelihood, impact, score, and assigned owner.

Risk #	Date	The Risk of/That	Caused by	Resulting In	Consequence/Impact	Likelihood Score	Impact Score	Risk Score	Priority (H, M, L)	Mitigation	Risk Owner	Open/Clos
1	03-03-2026	Cybersecurity breach or ransomware event disrupts RPM monitoring and exposes patient data	Expansion of IoT-connected RPM devices, cloud-based vendor platforms, and increased endpoints (attack surface) in the RPM ecosystem	Service downtime, delayed clinical escalation, potential privacy breach, reputational damage and reduced RPM adoption	Operational disruption to monitoring workflows; legal/compliance exposure under BC/Canadian privacy requirements; recovery/remediation costs; loss of patient/provider trust and competitive disadvantage versus private digital health providers	3	5	15	H	Zero-trust access controls (MFA/least privilege), encryption in transit/at rest, continuous vulnerability management & patch SLAs, device/vendor security reviews, ransomware playbook + backups and incident response drills, staff phishing training	Chief Information Security Officer (CISO) / Digital Health	Open
3	03-03-2026	Opportunity to strengthen Fraser Health's digital health leadership through rapid scaling of RPM	Rising demand for remote monitoring and chronic care support, plus system pressure to shift care from hospital to home	Higher RPM adoption, improved outcomes, and efficiency gains—creating competitive advantage in BC's digital health ecosystem	Operational: fewer avoidable admissions and better capacity management; financial: improved resource efficiency and stronger case for sustained funding; strategic: enhanced reputation as a digital health leader and higher stakeholder confidence	4	5	20	H	Scale eligibility cohorts and enrollment capacity, secure device supply/vendor SLAs, integrate RPM into care pathways/EMR, invest in analytics for early deterioration detection, track equity access (rural/underserved) and adjust deployment	Chief Medical Information Officer (CMIO) / Digital Innovation	Open
5	07-03-2026	Reliance on a single RPM vendor limits Fraser Health's ability to adopt more competitive solutions	Vendor lock-in due to long-term contracts and proprietary systems	Reduced flexibility to integrate new technologies or switch vendors	Higher operational costs and slower response to market innovation	3	3	9	M	Use multi-vendor strategies, negotiate flexible contracts, and adopt interoperable standards for RPM platforms	Procurement & Digital Health Technology Team	Open

Risk 1: Cybersecurity Breach

The risk of a cybersecurity breach or ransomware event disrupting RPM monitoring services and exposing patient data, caused by the expansion of IoT-connected RPM devices, cloud-based vendor platforms and increased digital endpoints in the RPM ecosystem, resulting in service downtime, delayed clinical escalation, privacy breaches, reputational damage and reduced RPM adoption.

Healthcare organizations are among the most targeted sectors for

cyberattacks globally due to the high value of patient data and the pressure placed on health systems to resolve disruptions quickly to protect patient safety (Gordon et al., 2020). Expanding the RPM ecosystem significantly increases the attack surface available to malicious actors, introducing new vulnerabilities through connected devices, vendor platforms, and increased data transmission volumes. Fraser Health would face legal and regulatory consequences under the British Columbia Personal Information Protection Act (PIPA) and Canadian federal privacy legislation if patient data were exposed, in addition to significant financial recovery costs and reputational damage.

Risk 2: System Integration Failure with Existing Healthcare Systems

The risk of RPM platform failing to integrate seamlessly with Fraser Health's existing Electronic Health Records (EHR), Laboratory Information Systems (LIS), and clinical workflow management systems caused by technical incompatibilities, inadequate interoperability standards, vendor platform limitations, insufficient integration testing, and the complexity of legacy healthcare IT infrastructure, resulting in data silos, duplicated data entry, delayed clinical decision-making, workflow disruption, provider frustration, reduced RPM platform adoption, and compromised patient safety.

Healthcare IT environments are notoriously complex, with legacy systems, multiple vendors, and varied data standards creating significant integration challenges. The RPM platform must exchange data bidirectionally with existing systems, vital signs must flow automatically into patient records, and alerts must trigger appropriate

clinical responses. Integration failures create cascading problems: workflow disruption from toggling between systems, data integrity risks from manual entry, and delayed interventions when alerts fail to reach clinicians. Healthcare IT projects historically experience delays, cost overruns, and functionality gaps, often leading to provider resistance and platform abandonment. Fraser Health would face extended costs, productivity losses, contract penalties and critical safety risks if clinical information fails to flow effectively between RPM and existing care delivery systems.

Risk 3: Regulatory Non-Compliance

The risk of Fraser Health's RPM platform failing to meet regulatory requirements under the British Columbia Personal Information Protection Act (PIPA), federal Personal Information Protection and Electronic Documents Act (PIPEDA), and health data management regulations, caused by inadequate privacy controls, insufficient consent management, improper data retention practices, weak vendor compliance oversight, gaps in security audit trails, and the complexity of managing continuous remote health data collection, resulting in regulatory investigations, mandatory breach notifications, financial penalties up to $500,000, mandatory service suspension, legal liability, class-action lawsuits and severe long-term reputational damage.

Healthcare organizations in British Columbia operate under strict privacy frameworks, with RPM platforms processing sensitive personal health information continuously across data collection, storage, use and retention. The regulatory landscape is complex, with PIPA governing personal information use, PIPEDA applying to cross-provincial data

flows, and the BC Office of the Information and Privacy Commissioner providing health-specific directives. Fraser Health remains legally accountable even when third-party vendors process patient data, amplifying compliance risks. Non-compliance consequences are severe: the OIPC can conduct investigations and impose financial penalties, patients can pursue lawsuits, and the provincial health ministry may mandate service suspension. Beyond legal penalties, privacy breaches create lasting reputational damage that erodes public trust and undermines patient willingness to participate in digital health initiatives for years following an incident.

3. Qualitative Risk Assessment

Qualitative risk assessment was conducted using two primary analytics-based methodologies: Scenario Analytics and Industry Fusion Analytics. These techniques enabled the team to evaluate each risk's potential severity and likelihood of occurrence without requiring fully quantified data, providing a robust foundation for subsequent quantitative analysis.

3.1 Methodology: Scenario Analytics

Scenario Analytics was applied by constructing plausible future scenarios related to each risk, evaluating how different combinations of internal and external factors could interact to produce risk outcomes. For the Cybersecurity Breach risk, scenarios ranged from ransomware attacks on vendor infrastructure affecting thousands of patients, to

insider threats exfiltrating data to Advanced Persistent Threats manipulating IoT device readings. For System Integration Failure risk, scenarios explored incompatible data formats requiring manual entry, six-month integration delays causing budget overruns, and failed alert delivery resulting in delayed interventions. For Regulatory Non-Compliance risk, scenarios examined vendors storing data on U.S. servers triggering OIPC investigations, inadequate consent processes leading to service suspension and high-profile privacy breaches creating national reputational damage.

3.2 Methodology: Industry Fusion Analytics

Industry Fusion Analytics was applied by drawing insights from analogous digital health implementations in comparable healthcare systems. Evidence from Canadian healthcare organizations revealed that cybersecurity incidents have increased significantly, with ransomware attacks affecting major systems including Toronto's University Health Network and Alberta Health Services causing service disruptions and data exposure (Cyber Security Centre of Canada, 2024). System integration challenges were assessed by examining Ontario Health's digital health projects, which experienced 40% budget overruns and 8-month delays due to EHR interoperability issues, and the UK's NHS integration failures requiring extensive remediation. Regulatory non-compliance risks were informed by documented breaches including LifeLabs' 2019 incident affecting 15 million Canadians resulting in class-action lawsuits and OIPC investigations, and Quebec health authority violations leading to service suspensions and multi-million-dollar penalties. This cross-industry analysis

informed the likelihood and impact ratings applied to each risk.

3.3 Risk Calculation and Heat Map Results

Risk Calculation

Probability of Occurrence	
Very Unlikely	1
Somewhat unlikely	3
50-50 possibility	5
Somewhat likely	7
Very likely	9

Impact of Risk		Financial	Operational	Competitive Risks	Reputational	Technical	Legal/Compliance
Very Low	1			Micro-Competitor Market Entry	Individual Negative Comments	UI/UX Cosmetic Changes	Administrative Paperwork
Somewhat Low	2	Device Cost Fluctuations, Administrative Overhead Increases	Minor Scheduling Delays, Equipment Delivery Delays	Minor Competitor Feature Updates	Minor Media Mentions	Minor Software Updates	Minor Compliance Updates
Moderate	4	Revenue Model Disadvantage, Lost Opportunity Costs		Talent Acquisition Disadvantage	High-Visibility Accountability		Data Sovereignty Requirements
Somewhat High	6	Venture Capital Advantage, High Switching Costs, Unexpected Costs for Remote Monitoring Infrastructure, Budget Reallocation Away from RPM Programs	Scale Complexity, Organizational Agility, Hospital Workflow Disruption During RPM Deployment, Insufficient Staff Training on Remote Monitoring Systems	Dominant Competitor Scale, Proven Clinical Outcomes, Mass Competitor Market Exit, New Digital Health Entrants Targeting Remote Care Services, Private Healthcare Providers Offering Faster Remote	Innovation Perception Gap, Social Media Amplification, Trust Paradox, Negative Patient Experience with Remote Monitoring Devices	Consumer Device Integration, Legacy System Constraints, Device Connectivity Failures During Patient Monitoring, Software Integration Issues Between Monitoring Devices and Hospital Systems	Vendor Lock-In Economics, Information Blocking Practices, Procurement Speed Disadvantage
High	8	Market Growth Explosion	24/7 Availability Advantage, Enrollment Friction, Delayed Clinical Response to Remote Monitoring Alerts	Market Saturation, Integrated Service Offerings, Partnerships Between Technology Firms and Private Clinics	Consumer Expectation Shift, Public Concerns About Accuracy of Remote Monitoring Data	Efficiency Revolution, Interoperability Barrier	Regulatory Penalties, Complete Privacy Law Elimination
Extremely High	9	Non-Compliance with Provincial Health Data Regulations	Access Speed Gap, Complete System Outage		Complete Public Rejection	AI Capability Gap, Technology Regression	

Risk #	The Risk of/That	Likelihood	Impact Score
1	Market Growth Explosion	9	8
2	Venture Capital Advantage	7	6
3	High Switching Costs	7	6
4	Revenue Model Disadvantage	5	4
5	Lost Opportunity Costs	3	4
6	Device Cost Fluctuations	5	2
7	Administrative Overhead Increases	3	2
8	Unexpected Costs for Remote Monitoring Infrastructure	5	6
9	Budget Reallocation Away from RPM Programs	3	6
10	Access Speed Gap	9	9
11	24/7 Availability Advantage	9	8
12	Scale Complexity	9	6
13	Organizational Agility	7	6
14	Enrollment Friction	7	8
15	Minor Scheduling Delays	5	2
16	Equipment Delivery Delays	3	2
17	Complete System Outage	1	9
18	Hospital Workflow Disruption During RPM Deployment	7	6
19	Insufficient Staff Training on Remote Monitoring Systems	5	6
20	Delayed Clinical Response to Remote Monitoring Alerts	5	8
21	Market Saturation	9	8
22	Dominant Competitor Scale	9	6
23	Integrated Service Offerings	7	8
24	Proven Clinical Outcomes	5	6
25	Talent Acquisition Disadvantage	5	4

26	Minor Competitor Feature Updates	7	2
27	Micro-Competitor Market Entry	5	1
28	Mass Competitor Market Exit	1	6
29	New Digital Health Entrants Targeting Remote Care Services	7	6
30	Private Healthcare Providers Offering Faster Remote Monitoring	5	6
31	Partnerships Between Technology Firms and Private Clinics	5	8
32	Consumer Expectation Shift	9	8
33	Innovation Perception Gap	7	6
34	High-Visibility Accountability	9	4
35	Social Media Amplification	7	6
36	Trust Paradox	3	6
37	Minor Media Mentions	5	2
38	Individual Negative Comments	7	1
39	Complete Public Rejection	1	9
40	Negative Patient Experience with Remote Monitoring Devices	5	6
41	Public Concerns About Accuracy of Remote Monitoring Data	5	8
42	AI Capability Gap	9	9
43	Efficiency Revolution	9	8
44	Consumer Device Integration	7	6
45	Interoperability Barrier	9	8
46	Legacy System Constraints	5	6
47	Minor Software Updates	7	2
48	UI/UX Cosmetic Changes	5	1
49	Technology Regression	1	9
50	Device Connectivity Failures During Patient Monitoring	7	6
51	Software Integration Issues Between Monitoring Devices and Hospital Systems	5	6
52	Vendor Lock-In Economics	9	6
53	Information Blocking Practices	7	6
54	Regulatory Penalties	3	8
55	Procurement Speed Disadvantage	9	6
56	Data Sovereignty Requirements	9	4
57	Minor Compliance Updates	5	2
58	Administrative Paperwork Changes	3	1
59	Complete Privacy Law Elimination	1	8
60	Non-Compliance with Provincial Health Data Regulations	5	9

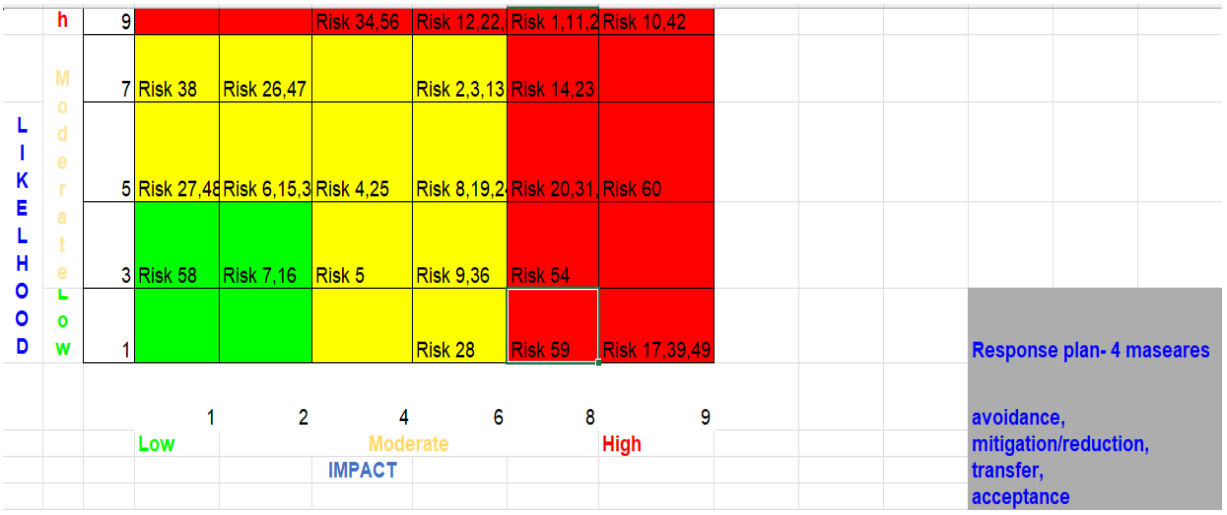
	Likelyhood/Impact	1	2	4	6	8	9
	1				1	1	3
	3	1	2	1	2	1	
	5	2	4	2	7	3	1
	7	1	2		10	2	
	9			2	4	6	2

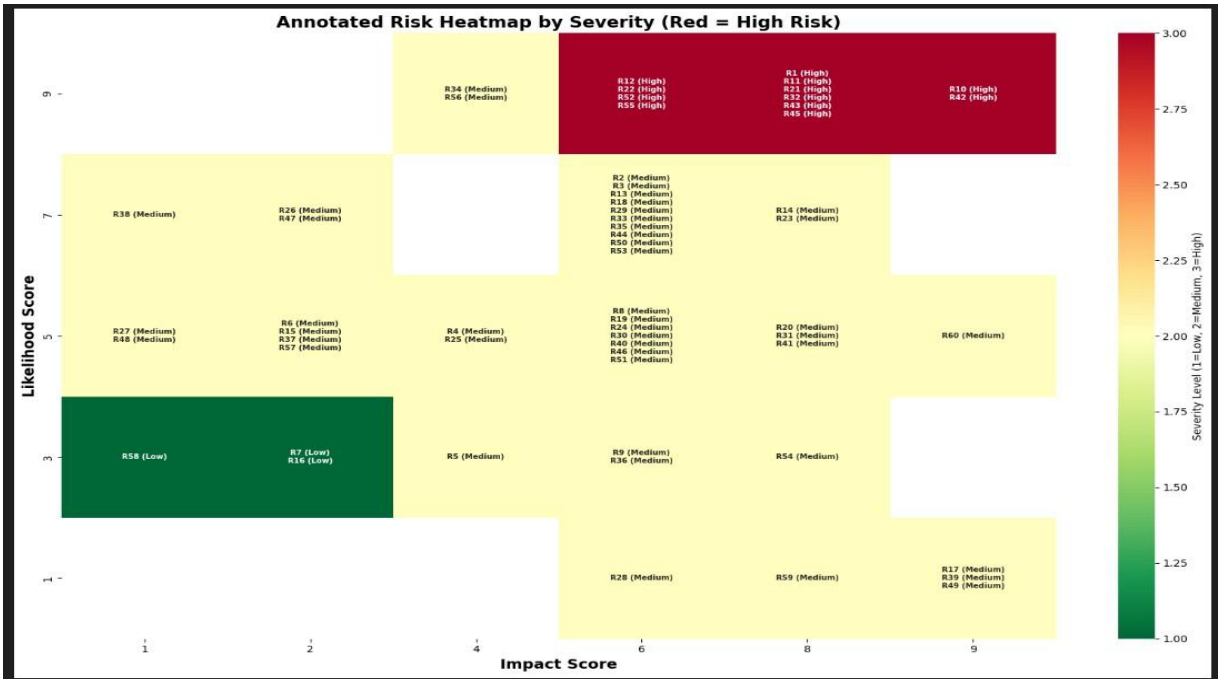
	Likelyhood/Impact	Low Impact	Moderate Impact	High Impact
Total risk score	Unlikely	Low Risk (3)		
	Likely		Moderate Risk (32)	
	Highly likely			High Risk (25)

Following qualitative scenario analysis, each risk was evaluated using numerical likelihood and impact scores applied to the Risk Calculation Sheet. The scoring matrix uses a scale where likelihood is rated as: Very Unlikely (1), Somewhat Unlikely (3), 50-50 Possibility (5), Somewhat Likely (7), and Very Likely (9) and the impact across the financial,

operational, competitive, reputational, technical and legal/compliance dimensions.

Risk Heatmap

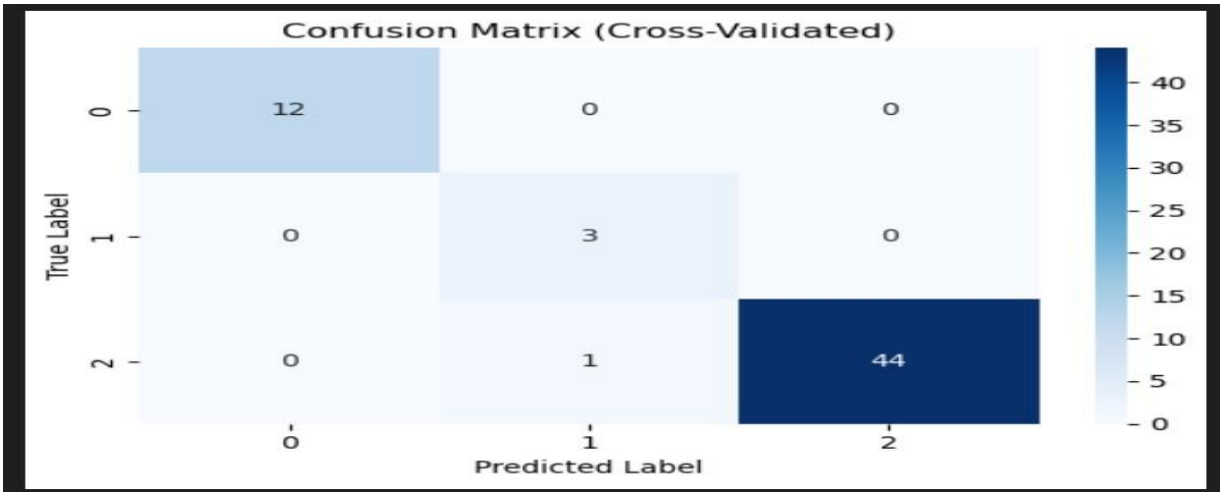




The Probability-Impact Heat Map visualized risk severity across all 60 identified risk factors. The most critical risks including Market Growth Explosion (R-1), Access Speed Gap (R-10), AI Capability Gap (R-42), and 24/7 Availability Advantage (R-11) clustered in the upper-right quadrant (likelihood 9, impact 8–9), indicating the high-severity areas

requiring immediate management attention. Cybersecurity-related risks, system integration concerns and regulatory compliance risks consistently appeared in the moderate-to-high severity regions of the heat map.

Cross Validation Confusion Matrix



The cross-validated confusion matrix produced by the Random Forest classification model confirmed high accuracy in distinguishing risk severity levels, with 12 correctly classified Low-risk observations, 3 Medium-risk and 44 High-risk observations with only one misclassification across the full dataset. This validated the qualitative scoring methodology.

3.4 Qualitative Assessment Conclusions by Risk

Risk 1 - Cybersecurity Breach: Qualitative analysis confirmed this risk as HIGH priority. The expansion of IoT endpoints and cloud-based vendor integration creates systemic vulnerability that scenario analysis identified as likely to materialize within the 18–24 months of full

deployment without robust controls. Industry fusion analytics demonstrated that healthcare organizations face increasing targeted attacks, with ransomware representing the dominant threat vector affecting comparable Canadian health systems. This risk was identified as a candidate for further quantitative analysis using I&W techniques.

Risk 2 - System Integration Failure: Qualitative assessment rated this risk as MEDIUM-HIGH priority. Scenario analytics revealed multiple failure pathways ranging from technical incompatibilities to workflow disruption, while industry fusion analytics demonstrated that healthcare IT integration projects consistently experience significant delays and cost overruns. The complexity of Fraser Health's existing clinical information infrastructure, combined with the bidirectional data exchange requirements of the RPM platform, creates substantial integration challenges. This risk requires rigorous integration testing protocols and was elevated for quantitative analysis using Monte Carlo simulation to model financial impact distributions.

Risk 3 - Regulatory Non-Compliance: Qualitative analysis confirmed this risk as HIGH priority with potentially critical consequences. The complex regulatory landscape spanning PIPA, PIPEDA and OIPC directives, combined with continuous processing of sensitive personal health information through third-party vendors, creates substantial compliance exposure. Industry fusion analytics revealed documented cases of major privacy breaches at Canadian healthcare organizations resulting in severe legal, financial and reputational consequences. The

potential for mandatory service suspension and long-term erosion of public trust elevates this risk to immediate attention priority, warranting detailed quantitative assessment through I&W analysis and Monte Carlo financial impact modelling.

4. Quantitative Risk Assessment

The quantitative risk assessment was conducted using the Indicators and Warning (I&W) analysis technique, supplemented by Monte Carlo simulation for financial impact modeling and Random Forest machine learning for independent severity classification. These complementary methods transformed the qualitative risk evaluation into a continuous, data-driven monitoring framework with clear early warning signals.

4.1 Indicators and Warning (I&W) Analysis

Indicators & Warnings (I&W) Risk Classification

Risk	Indicator	Unit	Low Boundary	Moderate Boundary	Current Value	Direction	Risk Score	Risk Level	Action Priority	Interpretation
Cybersecurity Breach	Intrusion Attempts	# per week	5	15	18	Higher is Worse	3	High	Immediate Action	High exposure to cyber threats and attempted unauthorized
Cybersecurity Breach	Vulnerability Score	1–10 scale	4	7	8	Higher is Worse	3	High	Immediate Action	System security weaknesses remain above the acceptable
Cybersecurity Breach	Data Encryption	%	95.0	85.0	90.0	Lower is Worse	2	Moderate	Review	Encryption coverage should improve to reduce breach
System Integration Failure	System Downtime	Hours/month	3	6	7	Higher is Worse	3	High	Immediate Action	Operational disruption may affect continuity of patient
System Integration Failure	Error Rate	%	5.0	10.0	12.0	Higher is Worse	3	High	Immediate Action	Integration instability may interrupt clinical workflows.
System Integration Failure	API Response Time	ms	200	300	320	Higher is Worse	3	High	Immediate Action	Slow performance can delay data exchange and clinical
Regulatory Non-Compliance	Audit Findings	# per quarter	1	3	4	Higher is Worse	3	High	Immediate Action	Audit findings above tolerance may trigger penalties or
Regulatory Non-Compliance	Staff Training	%	95.0	85.0	88.0	Lower is Worse	2	Moderate	Review	Training coverage is below the ideal level for privacy

d privacy

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  SECTION 1 — I&W COMPOSITE SCORES
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                 Risk     Mean  Total  Count Overall_Level  Action_Priority
  Cybersecurity Breach 2.666667      8      3          High Immediate Action
Regulatory Compliance 2.666667      8      3          High Immediate Action
   System Integration 3.000000      9      3          High Immediate Action
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Risk	Mean	Total	Count	Overall_Level	Action_Priority
Cybersecurity Breach	2.666667	8	3	High	Immediate Action
Regulatory Compliance	2.666667	8	3	High	Immediate Action
System Integration	3	9	3	High	Immediate Action

I&W Risk Heatmap

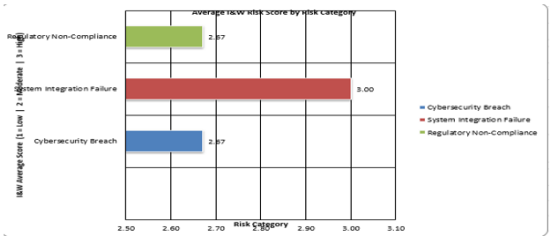
The I&W Risk Table sheet.

Total Indicators	High Indicators	Moderate Indicators	Low Indicators
9	7	2	0

Risk	Average Score	Overall Level	High Indicators
Cybersecurity Breach	2.7	High	2
System Integration Failure	3.0	High	3
Regulatory Non-Compliance	2.7	High	2

Indicator Heatmap (1 = Low, 2 = Moderate, 3 = High)

Risk	Indicator 1	Indicator 2	Indicator 3
Cybersecurity	3	3	2
System Integration	3	3	3
Regulatory	3	2	3



Risk	Indicator 1	Indicator 2	Indicator 3
Cybersecurity	Intrusion Attempts	Vulnerability Score	Data Encryption
System Integration	System Downtime	Error Rate	API Response Time
Regulatory	Audit Findings	Staff Training	Data Incidents

The I&W analysis revealed that 7 of 9 indicators across the three risk areas fell in the HIGH category, with only 2 indicators (Data Encryption coverage at 90% and Staff Training coverage at 88%) rated as Moderate. This data provides compelling quantitative evidence that all three risk areas require immediate remediation action before full RPM platform deployment. Composite I&W scores were Cybersecurity Breach (2.67/3.0), System Integration Failure (3.0/3.0) and Regulatory Non-Compliance (2.67/3.0), all classified as High risk with Immediate Action priority.

4.2 Monte Carlo Simulation – Financial Impact Modeling

Risk	Mean_CAD_M	P95_CAD_M	P5_CAD_M	StdDev_CAD_M	Prob_Over_10M
Cybersecurity Breach	7.78	12.07	4.09	2.4	20
System Integration Failure	6.13	9.37	3.29	1.86	2.3
Regulatory Non-Compliance	4.91	8.4	1.4	2.08	0.7

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  SECTION 2 — MONTE CARLO RESULTS (CAD Millions)
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                    Risk  Mean_CAD_M  P95_CAD_M  P5_CAD_M  StdDev_CAD_M  Prob_Over_10M
      Cybersecurity Breach        7.78      12.07      4.09          2.40           20.0
System Integration Failure        6.13       9.37      3.29          1.86            2.3
 Regulatory Non-Compliance        4.91       8.40      1.40          2.08            0.7
```

The Monte Carlo results are particularly significant for the Cybersecurity Breach risk where the mean financial impact of $7.78M CAD and 95th percentile impact of $12.07M CAD with a 20% probability of exceeding $10M CAD represents the substantial financial exposure to a publicly funded health authority. System Integration Failure carries a mean impact of $6.13M CAD, representing costs associated with remediation, care continuity disruptions and vendor contract penalties. Regulatory non-compliance with a mean impact of $4.91M CAD, reflects potential regulatory penalties, legal costs and reputational remediation expenses.

4.3 Random Forest Machine Learning Classification

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  SECTION 3 — RANDOM FOREST PREDICTIONS
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                            likelihood  impact  iw_score Severity  Confidence
Cybersecurity Breach                 7       8       0.9     High    0.819778
System Integration Failure           5       6       1.0   Medium    0.807383
Regulatory Non-Compliance            5       9       0.9   Medium    0.635354
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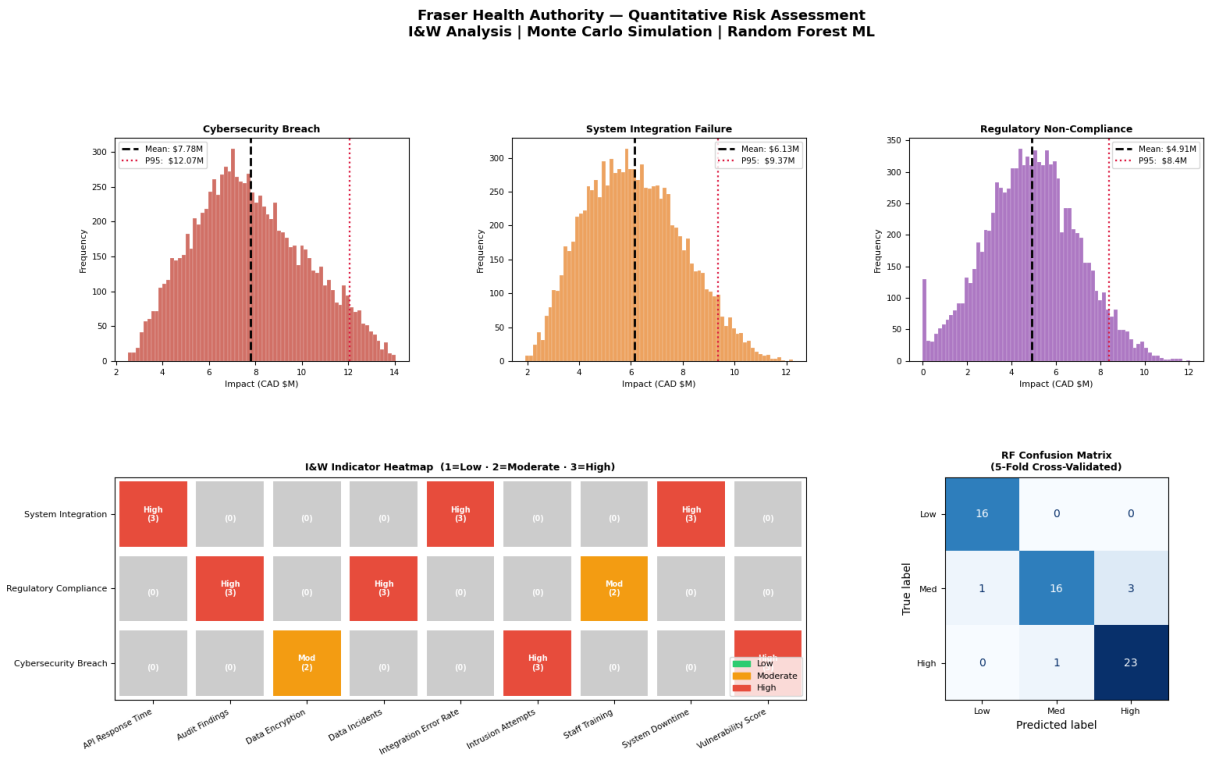
A Random Forest classifier with 5-fold cross-validation was applied to independently validate the I&W severity classifications. The model analyzed patterns in indicator data and predicted risk severity levels with the following results: Cybersecurity Breach was classified as High severity with 81.9% confidence; System Integration Failure as Medium severity with 80.7% confidence; and Regulatory Non-Compliance as Medium severity with 63.5% confidence.

The convergence between I&W composite scores (all classified as High/Immediate Action) and the ML predictions (High and Medium) is

noteworthy. The moderate ML confidence scores for System Integration and Regulatory risks reflect the greater uncertainty in those risk domains, precisely the areas where the Monte Carlo distribution shows wider variance. This convergence strengthens confidence in the overall risk assessment outcomes and supports the recommendation for immediate management action across all three quantitatively assessed risks.

4.4 Quantitative Risk Assessment Dashboard



Quantitative Assessment Summary — Fraser Health RPM

Risk	I&W Score	Mean Impact	P95 Impact	Std Dev	P(>$10M)	ML Class.	ML Conf.	Action
Cybersecurity Breach	2.7 / 3.0	$7.78M	$12.07M	$2.4M	20.0%	High	82%	Immediate
System Integration Failure	3.0 / 3.0	$6.13M	$9.37M	$1.86M	2.3%	Medium	81%	Immediate
Regulatory Non-Compliance	2.7 / 3.0	$4.91M	$8.4M	$2.08M	0.7%	Medium	64%	Immediate

The analysis indicates that cybersecurity breach is the most critical risk, with the highest expected impact and likelihood of extreme losses that

requiring top priority action. System integration failure and regulatory non-compliance are classified as moderate risks but remain operationally important due to key warning indicators. Overall, the analysis highlights the need for immediate, data-driven risk mitigation,

with a primary focus on cybersecurity while continuously monitoring integration and compliance risks.

5. Risk Response Strategy

Fraser Health's risk response strategy for the Remote Patient Monitoring (RPM) initiative is designed to align with the COSO Enterprise Risk Management framework, ensuring that risks are proactively managed while supporting strategic healthcare delivery goals. Based on qualitative and quantitative risk assessments, three key risks have been prioritized, each with a targeted response approach.

Risk 1 - Cybersecurity Breach (High Priority Risk)
Strategy: Mitigate and Transfer

Cybersecurity represents the most critical risk, with the highest likelihood and potential financial impact (mean ~$7.78M CAD). The expansion of IoT-enabled RPM devices and cloud-based platforms significantly increases the organization's attack surface.

Key Actions:

- Implement advanced security controls (encryption, MFA, Zero Trust Architecture).

- Establish continuous monitoring through a 24/7 Security Operations Center.
- Conduct regular vulnerability testing and staff training.
- Transfer residual financial risk through cybersecurity insurance and vendor accountability agreements.

Outcome: Reduced likelihood and impact of breaches with improved system resilience and regulatory compliance.

Risk 2 - System Integration Failure (Medium-High Priority Risk)

Strategy: Mitigate and Contingency Planning

System integration represents a significant operational risk, with potential financial impact (mean ~$6.13M CAD) and direct implications for clinical workflow efficiency and patient safety. The complexity of integrating RPM platforms with existing EHR, LIS and clinical systems requires proactive mitigation.

Key Actions:

- Conduct comprehensive integration architecture review with HL7 FHIR interoperability specialists.
- Conduct comprehensive integration architecture review with HL7 FHIR interoperability specialists.
- Implement phased rollout with pilot sites to identify integration issues early.

- Establish vendor Service Level Agreements (SLAs) for integration performance and maintain dedicated budget reserve ($1.5M) for extended integration services.

Outcome: Seamless bidirectional data exchange between RPM platform and existing clinical systems, minimized workflow disruption, and sustained provider adoption.

Risk 3 - Regulatory Non-Compliance (High Priority Risk)

Strategy: Mitigate and Avoid (Zero-Tolerance Approach)
Regulatory non-compliance represents a critical risk with potential financial impact (mean ~$4.91M CAD), but more significantly, the possibility of mandatory service suspension and severe long-term reputational damage. Fraser Health maintains legal accountability even when third-party vendors process patient data.

Key Actions:

- Establish RPM Privacy Oversight Committee and appoint dedicated RPM Privacy Coordinator.
- Implement privacy-by-design principles with comprehensive technical controls (encryption, DLP, geofenced Canadian data storage, audit logging).
- Require vendor compliance attestations, conduct annual vendor privacy audits, and establish data processing agreements with clear accountability.

- Mandate annual privacy training for all staff and implement layered patient consent processes with transparent data use disclosure.

Outcome: Full regulatory compliance with PIPA, PIPEDA and OIPC directives, maintained public trust, and zero major privacy breaches.

Strategic Prioritization

- **Immediate Action:** Cybersecurity risk mitigation and regulatory compliance framework establishment.
- **Pre-Implementation Focus:** System integration testing, architecture review, and vendor SLA agreements.
- **Ongoing Management:** Continuous privacy oversight, integration performance monitoring, and security posture assessment.
- **Zero-Tolerance Areas:** Regulatory non-compliance and cross-border data flows.

Continuous Monitoring and Governance

All risk responses are supported by:

- Key Risk Indicators (KRIs)
- Indicators & Warning (I&W) systems
- Machine learning-based risk monitoring

- Regular audits and performance reviews

Overall Strategy

Fraser Health's risk response strategy balances risk mitigation, operational excellence, and legal accountability by prioritizing cybersecurity, establishing robust integration testing, and implementing zero-tolerance privacy controls. This integrated approach ensures successful RPM platform deployment while protecting patient data, maintaining regulatory compliance, and delivering high-quality care through continuous monitoring and governance oversight.

6. Key Risk Indicators (KRI's) and their trigger points for action

Key Risk Indicators provide Fraser Health with real-time visibility into risk exposure levels and early warning signals that enable proactive intervention before risks materialize into significant adverse events. The KRI framework established below integrates findings from the I&W analysis, qualitative risk scoring and strategic priorities identified in the risk response strategy.

6.1 KRI Framework Structure

Each KRI is categorized using a three-tier threshold system:

- **GREEN (Acceptable)**: Risk exposure within tolerance; routine monitoring continues.

- **YELLOW (Caution)**: Risk exposure increasing; enhanced monitoring and preparatory actions required.
- **RED (Critical)**: Risk exposure exceeds tolerance; immediate intervention and escalation to executive leadership required.

6.2 Risk 1: Cybersecurity Breach - KRIs and Trigger Points

Risk 1: Cybersecurity Breach - KRIs and Trigger Points

KRI	Measurement Frequency	Green Threshold	Yellow Threshold	Red Threshold	Action Required
Unsuccessful External Attack Attempts	Daily	<50 attempts/day	50-150 attempts/day	>150 attempts/day	**RED**: Activate Incident Response Team; Conduct Immediate Threat Assessment; Notify CISO
Time to Peach Critical Vulnerabilities	Weekly	<48 hours	48-96 hours	>96 hours	**RED**: Suspend Affected Systems; Emergency Patching Protocol; Executive Briefing
Percentage of RPM Devices with Updated Security Firmware	Monthly	>=95%	85-94%	<85%	**YELLOW**: Accelerate Update Deployment; Identify Non-Compliant Devices. **RED**: Quarantine Outdated Devices; Mandatory Update Enforcement
Factor Authentication (MFA) Adoption Rate	Monthly	>=98%	90-97%	<90%	**RED**: Mandate MFA for All Users; Disable Non-MFA Access
Training Completion Rate	Quarterly	>=95%	85-94%	<85%	**YELLOW**: Targeted Training Campaigns. **RED**: Mandatory Completion; Access Restrictions for Non-Compliant Staff
Privileged Access Account Reviews Completed	Quarterly	100% on schedule	1-15 days overdue	>15 days overdue	**RED**: Suspend Privileged Access Pending Review; CISO Escalation
Encryption Coverage (In-Transit and At-Rest)	Monthly	100%	95-99%	<95%	**YELLOW**: Identify Unencrypted Data Flows; Remediation Plan. **RED**: Immediate Encryption Deployment; Suspend non-compliant Data Transfers
Security Assessment Currency	Quarterly	Within 90 days	91-180days	>180 days	**RED**: Suspend Vendor Data Processing; Emergency Security Audit
Security Incident Response Time	Per Incident	<2 hours	2-6 hours	>6 hours	**RED**: Activate Crisis Management; External Cybersecurity Support; Board Notification

Composite Cybersecurity Risk Score: Average of all 9 KRI severity levels.

Trigger Point: Composite score >=2.0 requires immediate executive briefing and accelerated mitigation.

6.3 Risk 2: System Integration Failure - KRIs and Trigger Points

Risk 2: System Integration Failure - KRIs and Trigger Points

KRI	Measurement Frequency	Green Threshold	Yellow Threshold	Red Threshold	Action Required
Bidirectional Data Synchronization Success Rate	Daily	>=99.5%	97-99.4%	<97%	**YELLOW**: Root Cause Analysis; Vendor Engagement. **RED**: Halt New Patient Enrollments; Emergency Integration Review
RPM Alert Delivery Failure Rate to Clinical Systems	Daily	<0.5%	0.5-2%	>2%	**RED**: Manual Alert Monitoring Protocol; Immediate Vendor Escalation; Clinical Leadership Notification
Average Data Latency (RPM Device to EHR)	Hourly	<5 minutes	5-15 minutes	>15 minutes	**YELLOW**: Performance Optimization. **RED**: Clinical Safety Review; Consider System Rollback
Manual Data Entry Incidents per 1,000 Encounters	Weekly	<10	10 to 30	>30	**YELLOW**: Workflow Analysis; Integration Gap Identification. **RED**: Integration Failure Declared; Contingency Plan Activation
Integration Testing Pass Rate	Per Release	>=98%	90-97%	<90%	**RED**: Halt Deployment; Comprehensive Regression Testing; Vendor Remediation
Provider Satisfaction Score (Integration Usability)	Monthly	>=4.0/5.0	3.0-3.9/5.0	<3.0/5.0	**YELLOW**: Workflow Support Enhancement. **RED**: Integration Redesign Required; User Acceptance Testing
System Downtime (Unplanned, RPM Platform)	Monthly	<1 hour/month	1-4 hours/month	>4 hours/month	**RED**: Vendor SLA Breach; Financial Penalties; Redundancy Review
Integration Budget Variance	Monthly	<=10% over budget	10-25% over budget	>25% over budget	**YELLOW**: Budget Reallocation Planning. **RED**: Executive Approval Required; Project Scope Review
Open Integration Defects (Critical/High Severity)	Weekly	<=5	6 to 15	>15	**RED**: Defect Resolution Sprint; Vendor Escalation; Deployment Hold

Composite Cybersecurity Risk Score: Average of all 9 KRI severity levels.

Trigger Point: Composite score >=2.0 requires CIO review and integration governance board meeting within 48 hours.

6.4 Risk 2: Regulatory Non-Compliance - KRIs and Trigger Points

Risk 3: Regulatory Non-Compliance - KRIs and Trigger Points

KRI	Measurement Frequency	Green Threshold	Yellow Threshold	Red Threshold	Action Required
Privacy Breach Incidents	Monthly	0 incidents	1-2 minor incidents	>=3 incidents OR 1 major incident	**YELLOW:** Breach Investigation; OIPC Notification Assessment. **RED:** Mandatory OIPC Reporting; Legal Counsel Engagement; Crisis Communications
Patient Consent Compliance Rate	Monthly	>=99%	95-98%	<95%	**YELLOW:** Consent Process Audit. **RED:** Suspend New Enrollments; Consent Remediation Program
Vendor Compliance Audit Currency	Quarterly	Audited within 12 months	13-18 months	>18 months	**YELLOW:** Schedule Emergency Audits. **RED:** Suspend Vendor Data Processing Pending Audit
Data Retention Policy Violations	Monthly	0 violations	1-3 violations	>3 violations	**YELLOW:** Data Purge Protocol Review. **RED:** Immediate Data Deletion; Policy Enforcement Overhaul
Cross-Border Data Transfer Incidents	Monthly	0 incidents	1 incident	>1 incident	**RED:** Immediate Data Repatriation; OIPC Notification; Vendor Contract Review
Privacy Training Completion Rate	Quarterly	>=95%	85-94%	<85%	**YELLOW:** Targeted Training. **RED:** Mandatory Completion; Access Suspension for non-compliant Staff
Security Audit Trail Coverage	Weekly	100% of systems	95-99%	<95%	**YELLOW:** Audit Logging Enhancement. **RED:** Compliance Breach; Immediate Remediation
Privacy Impact Assessment (PIA) Currency	Annually	Current within 12 months	13-18 months overdue	>18 months overdue	**RED:** Suspend RPM Operations Pending Updated PIA; OIPC Consultation
Regulatory Inquiry Response Time	Per Inquiry	<5 business days	5-10 business days	>10 business days	**RED:** Legal Escalation; Executive Oversight; Regulatory Relationship Management

Composite Cybersecurity Risk Score: Average of all 9 KRI severity levels.

Trigger Point: Composite score >=1.5 requires Privacy Officer escalation and immediate remediation planning. Score >=2.0 requires CEO notification.

6.5 KRI Monitoring and Governance Framework

Monitoring Responsibilities

- **Daily KRIs**: Monitored by RPM Operations Team; Automated dashboard alerts.
- **Weekly KRIs**: Reviewed by Risk Management Office; Reported to CIO/CISO.
- **Monthly KRIs**: Compiled in monthly Risk Dashboard; presented to Executive Risk Committee.

- **Quarterly KRIs**: Integrated into quarterly Board Risk Report.

Escalation Protocol

- **Any RED threshold breach**: Immediate notification to assigned risk owner + Risk Management Office.
- **Multiple YELLOW thresholds (≥3 in same risk category)**: Risk owner develops action plan within 48 hours.
- **Composite risk score triggers**: Convene emergency risk review session within 24 hours.

KRI Review Cycle

- KRI thresholds reviewed quarterly and adjusted based on operational experience.
- New KRIs added as risks evolve or new threats emerge.
- Annual comprehensive KRI framework validation aligned with strategic planning cycle.

Technology Enablement

- Automated KRI dashboard with real-time data feeds from security systems, EHR, vendor platforms.
- Machine learning-based anomaly detection to identify emerging risk patterns.
- Mobile executive dashboard for leadership visibility.

6.6 Integration with Risk Response Strategy

The KRI framework directly supports the risk response strategies outlined in Section 5:

- **Cybersecurity Mitigation**: KRIs provide continuous monitoring of security posture, enabling proactive threat detection and rapid response.
- **Integration Testing**: KRIs ensure seamless technical performance and clinical workflow effectiveness.
- **Regulatory Compliance**: KRIs enforce zero-tolerance approach to privacy violations and maintain legal accountability.

Continuous Improvement

KRI performance data feeds back into the ERM cycle, informing:

- Risk appetite refinement
- Control effectiveness assessment
- Strategic decision-making for RPM platform scaling

This comprehensive KRI framework ensures Fraser Health maintains proactive risk management throughout the RPM platform lifecycle, protecting patient safety, organizational reputation and strategic healthcare delivery objectives.

7. Recommendation

Fraser Health must implement three critical pre-deployment requirements before authorizing full RPM platform launch: (1) Establish comprehensive cybersecurity controls that include 24/7

Security Operations Center, Zero Trust Architecture, end-to-end encryption, and $10M CAD cybersecurity insurance coverage; (2) Create a robust regulatory compliance framework with a dedicated Privacy Oversight Committee, Privacy Coordinator, privacy-by-design principles, geofenced Canadian data storage, and annual vendor privacy audits; and (3) Conduct extensive integration testing with HL7 FHIR specialists, establish vendor Service Level Agreements, implement phased rollout beginning with pilot sites and allocate budget reserves totaling $16.5M CAD for integration remediation, cybersecurity incidents and compliance costs.

Following deployment, Fraser Health must maintain continuous risk monitoring through an automated KRI dashboard with real-time data feeds, machine learning-based anomaly detection and three-tier threshold alerts (GREEN/YELLOW/RED) supported by structured monitoring cadence from daily operational tracking through quarterly Board-level reporting. Implement strict escalation protocols requiring immediate executive notification for RED threshold breaches and 24-hour emergency reviews for composite score triggers with quarterly KRI threshold reviews and annual ERM framework validation to refine risk models and address emerging domains.

8. Conclusion

This Enterprise Risk Management assessment evaluates Fraser Health Authority's Remote Patient Monitoring (RPM) platform acquisition for 1.9 million residents using the COSO ERM framework across five progressive modules. Through integrated qualitative and quantitative methodologies including SWOT analysis, Scenario Analytics, I&W analysis, Monte Carlo simulation and Random Forest machine learning, three critical risks were identified: Cybersecurity Breach (highest priority, $7.78M CAD mean impact), System Integration Failure (medium-high priority, $6.13M CAD mean impact) and Regulatory Non-Compliance (high priority, $4.91M CAD mean impact with potential mandatory service suspension).

Quantitative analysis demonstrated that immediate action is required across all three risk domains, with I&W analysis showing 7 of 9 indicators in the HIGH category and composite risk scores of 2.67-3.0 out of 3.0. Monte Carlo simulations revealed substantial financial exposure including a 20% probability of cybersecurity losses exceeding $10M CAD, while machine learning validation confirmed severity assessments with 63.5-81.9% confidence levels. The convergence of findings across multiple analytical methods validates the need for proactive management intervention.

The recommended risk response strategy balances mitigation, operational excellence, and legal accountability through three core actions: implementing advanced cybersecurity controls with 24/7 monitoring and insurance-based risk transfer; conducting comprehensive integration testing with HL7 FHIR specialists and phased rollout; and establishing zero-tolerance privacy controls via a dedicated Privacy Oversight Committee with privacy-by-design principles. Fraser Health should proceed with RPM platform acquisition contingent upon immediate implementation of these frameworks, supported by a comprehensive KRI system with automated threshold alerts, as disciplined risk management execution enables transformative patient care benefits while maintaining acceptable risk exposure and protecting organizational reputation.

Appendix 1: Complete Risk Treatment and Response Plan (RT&RP)

1.Risk Overview

Fraser Health's integrated risk treatment strategy addresses three critical interdependent risks; Cybersecurity Breach (mean exposure $7.78M CAD), System Integration Failure (mean exposure $6.13M CAD), and Regulatory Non-Compliance (mean exposure $4.91M CAD) with combined exposure of $18.82M mean and $30.25M worst-case

scenario (95th percentile).

2.Investment Requirements

The comprehensive risk mitigation framework requires $15.1M Year 1 investment covering cybersecurity controls ($5.7M), privacy and compliance framework ($1.8M), system integration architecture and testing ($2.9M), and governance, training, and validation ($0.9M), with $5.5M ongoing annual commitment and $16.5M contingency reserves allocated for cybersecurity incidents ($10M), integration remediation ($1.5M), and regulatory compliance costs ($5M).

3. Pre-Deployment Requirements (Months 1-5)

Cybersecurity Controls: Implement 24/7 monitoring with fast threat detection, enforce Zero Trust with strong encryption, require widespread multi-factor authentication, deploy IoT and data protection systems, regularly test security, and maintain $10M in cyber insurance.

Privacy and Compliance Framework: Set up a privacy governance structure, complete required assessments and build the system with privacy-by-design. Ensure vendor compliance, keep all data within Canada, track PHI access and give patients clear control over their data through consent options and a rights portal.

System Integration Architecture: Hire HL7 FHIR experts to design a standards-based integration, assess IT infrastructure, map data between systems, and set up bidirectional data exchange. Test thoroughly (integration, performance for 10,000+ patients, and failover) to ensure high reliability, and establish vendor SLAs with clear metrics for uptime, latency, and error rates.

Governance and Training: Establish Enterprise Risk Committee with monthly oversight meetings, allocate $16.5M contingency reserves, complete mandatory cybersecurity and privacy training for all staff (>95% completion), conduct ransomware tabletop exercises with executive participation, and test all incident response plans.

4. Deployment Authorization Criteria (Month 6 - Go/No-Go Decision)

Fraser Health will approve RPM platform deployment only if strict technical, security, privacy and operational criteria are fully met including strong cybersecurity (24/7 monitoring, Zero Trust, encryption, MFA), resolved vulnerabilities, approved privacy measures, Canadian data residency, reliable system performance (high testing pass rates, low latency, >99% alert success), executed vendor agreements and completed staff training. If any requirement is unmet, deployment is delayed until all issues are resolved.

5. Pilot Deployment Phase (Months 6-9)

Run a pilot at 2–3 diverse sites with 100–200 patients to evaluate integration, workflows and user feedback while strictly monitoring security and privacy. Move to full deployment only if all targets are met, including very low error rates, no safety or security issues, high clinician (>85%) and patient (>90%) satisfaction, strong system performance and high SLA compliance (>95%).

6.Full Deployment Phase (Months 10-18)

Following successful pilot evaluation (Month 9 go/no-go decision),

execute phased expansion across all Fraser Health sites ($2M investment) with continuous KRI monitoring, quarterly vendor performance reviews and ongoing optimization based on operational experience and real-time data.

7. Ongoing Risk Monitoring Framework

Create an automated Key Risk Indicators dashboard tracking 18 metrics across cybersecurity, system integration and regulatory compliance, with GREEN/YELLOW/RED thresholds for proactive alerting. Monitor areas like security incidents, vulnerabilities, integration errors, data latency, alert delivery and privacy compliance.

Use a structured review cadence: daily monitoring with alerts, weekly risk reviews by leadership, monthly executive reporting and quarterly updates to the board, ensuring continuous oversight and timely response to risks.

8. Escalation Protocols

Define clear escalation rules: any RED threshold triggers immediate executive action, while multiple YELLOWs require a mitigation plan within 48 hours. High composite risk scores prompt urgent review meetings, with additional governance actions such as board, CEO, or integration board notifications within 24–48 hours depending on severity.

9. Contingency Plans

Cybersecurity Breach Response: Activate the incident response plan within 15 minutes, contain the issue, preserve evidence, and notify key stakeholders (privacy officer, legal). Assess the breach, report to regulators if required, involve insurers and forensic experts, communicate with affected parties and complete a post-incident review with corrective actions.

System Integration Failure Response: Within 24 hours, convene the Integration Governance Board, pause rollout beyond pilot sites, use a $1.5M contingency for vendor support, maintain care via manual workflows, escalate to vendor leadership, engage independent consultants, evaluate alternative vendors if unresolved in 90 days and inform executive leadership and the board of material risk.

Regulatory Non-Compliance Response: Immediately alert the Chief Privacy Officer and activate the Privacy Incident Response Plan. Within 24–72 hours, assess compliance gaps, notify regulators and affected patients if required, halt operations for material violations, engage legal counsel, develop and execute a remediation plan, inform leadership and the board, and implement enhanced monitoring and audits after remediation.

10. Success Metrics

Year 1 Targets: Zero successful cyberattacks or data breaches, zero OIPC investigations or regulatory penalties, zero patient safety events attributable to RPM, integration error rate <1%, clinical alert delivery >99%, clinician satisfaction >90%, patient satisfaction >90%, privacy training completion 100%, vendor SLA compliance >95%, and all KRI

composite scores maintained below critical threshold (<2.0).

Long-Term Indicators (Years 2-3): Hospital readmission reduction of 15-25% for RPM patients, improved chronic disease management outcomes, enhanced patient engagement scores, sustained regulatory compliance (zero violations), sustained cybersecurity posture (zero breaches), platform expansion to 10,000+ patients, and positive return on investment by Year 3 through reduced hospitalizations and improved care outcomes.

11. Continuous Improvement

Conduct quarterly KRI threshold reviews and adjustments based on operational experience, semi-annual risk assessment updates incorporating real-time operational data, annual comprehensive ERM framework validation aligned with strategic planning cycle, ongoing addition of new KRIs as risks evolve and emerging threats are identified, refinement of Monte Carlo models with actual operational data, and extension of analysis to emerging domains including AI integration, multi-vendor interoperability, and evolving regulatory requirements.

12. Final Authorization Recommendation

Fraser Health should acquire the RPM platform only after fully implementing the Integrated Risk Treatment and Response Plan. The financial and operational benefits justify the $15.1M Year 1 investment, as disciplined, coordinated risk management, covering compliance, security, and integration can control exposure and ensure safe, regulatory-compliant deployment. With proper execution, the platform

can deliver transformative remote care while safeguarding patient data and organizational reputation for 1.9 million residents.

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